

Convolutional Neural Network Pipeline with Feature Synthesis for Cardiovascular Risk Prediction

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Abstract—This work presents a convolutional neural network (CNN) pipeline for cardiovascular disease (CVD) classification using structured clinical data. The pipeline integrates outlier handling, clinically motivated feature engineering, and analysis of variance (ANOVA)-based feature selection. Three interaction features are introduced to capture age-related risk effects and blood pressure–cholesterol interactions. The CNN architecture is tailored for tabular inputs and comprises three one-dimensional convolutional layers with batch normalization and dropout. The model is evaluated on the Heart Disease dataset containing 1,025 patient records. On the test set, the proposed approach achieves 97.56% accuracy, a 97.63% F1-score, and an area under the receiver operating characteristic curve (AUC–ROC) of 99.80%. Sensitivity and specificity reach 98.10% and 97.00%, respectively, with five misclassifications out of 205 test samples. The training-testing accuracy gap is limited to 2.20%, indicating controlled overfitting. The full training process completes in 3.54 seconds. These results demonstrate that CNN-based models, when combined with targeted feature synthesis, can provide accurate and computationally efficient cardiovascular risk prediction from tabular clinical data.

Index Terms—Heart disease prediction, Convolutional neural networks, Medical diagnostics, Feature engineering, Healthcare artificial intelligence, Machine learning in medicine

I. INTRODUCTION

A. Motivation for Early Cardiovascular Risk Prediction

Cardiovascular disease remains a serious clinical problem. It affects large patient populations and places a sustained burden on healthcare systems. Despite improvements in treatment, many cases are still identified late. Preventive decision-making, therefore, remains difficult in routine clinical practice.

Traditional cardiovascular risk assessment relies on clinician-driven analysis and, in some cases, invasive diagnostic procedures. These approaches are costly and difficult to deploy at scale. In settings with limited clinical resources, this often leads to delays in assessment or incomplete follow-up. As a result, early detection is not consistent, and routine risk monitoring is not always performed in practice.

Deep learning (DL) models offer tools for addressing this problem. In particular, convolutional neural networks can model interactions between clinical variables rather than treating features independently. This study applies this capability to structured clinical data to support cardiovascular disease classification with practical computational cost.

B. Paper Organization and Structure

Related work on machine learning methods for CVD prediction is reviewed in Section II. Section III describes the dataset, preprocessing pipeline, feature engineering strategy, and the proposed 1D CNN model. Experimental results, evaluation metrics and limitations are presented in Section IV. Section V concludes with a summary and directions for future work.

II. RELATED WORK

CVD risk prediction has a long history in clinical research. Early work focused on statistical models, most commonly logistic regression and Cox proportional hazards analysis. Clinical risk scores such as the Framingham Risk Score, the Atherosclerotic Cardiovascular Disease (ASCVD) calculator, and QRISK variants are still used in practice. These tools are easy to use, but they rely on simplified assumptions. In particular, they often treat risk factors independently and do not account for complex interactions. This limits their effectiveness in more complex clinical settings, where reported discrimination typically remains moderate [1].

A large body of work has explored machine learning (ML) models as an alternative. Reported accuracy varies across studies. Efe et al. [2] explored feature selection using stability selection with a Hoeffding Tree classifier. Their reported accuracy for heart disease prediction was 84%. Anjum et al. [3] conducted a comparative study of six ML models for myocardial infarction prediction. In that study, Extreme Gradient Boosting (XGBoost) achieved the highest accuracy at 94.8%. Overall, these results show that preprocessing choices and model selection both affect performance.

Several studies report good performance from tree-based classifiers on structured cardiovascular datasets. Vallabhaneni et al. [4] combined feature ranking with Random Forest (RF). After addressing class imbalance using the Synthetic Minority Over-sampling Technique (SMOTE), accuracy reached 95.61%. Other studies report lower performance. Bhowmik et al. [5] compared logistic regression, RF, and support vector machine (SVM) models on a large dataset. The highest accuracy they reported was 72.33% using SVM. Logistic regression achieved the highest AUC–ROC of 0.781. This suggests variations in results across datasets.

Some studies focus on feature optimization rather than classifier design. Olawade et al. [6] applied Bald Eagle Search

Optimization (BESO) to select features before classification. They evaluated k-nearest neighbors (KNN), Logistic Regression, SVM, and RF. Reported accuracy reached 90% on the Framingham dataset and 92% on the Z-Alizadeh Sani dataset. That said, the feature-optimization step increases system complexity.

Some other studies have focused on specific patient cohorts or data settings. Sadiq et al. [7] examined dyslipidemia-related cardiovascular risk during pregnancy. Using RF models based on lipid profile variables, they reported accuracy up to 95%. Sianga et al. [8] incorporated geographical information alongside clinical features. Their best results were obtained using XGBoost, with accuracy of 95.24%. These results suggest that feature relevance can differ across populations.

Practical considerations for real-world deployment have also been examined. Kapila et al. [9] combined federated learning with feature selection and extraction methods, including linear discriminant analysis, reporting accuracies of 88.52% and 92.3% across benchmark datasets. Related issues were noted by Ahmed et al. [10], who demonstrated that careful preprocessing and feature selection can yield strong performance for heart failure survival prediction, with accuracy reaching 94%, even when advanced models are used.

In contrast to traditional ML approaches, the application of CNN-based models to structured clinical data has received limited attention. In tabular settings, deep architectures are sensitive to feature design and regularization choices, and performance gains are not guaranteed. This motivates approaches that combine clinically informed feature synthesis with CNN-based classification, with an emphasis on stable behavior and manageable computational cost.

III. METHODOLOGY

A. Dataset Description

The Heart Disease dataset was obtained from Kaggle. It contains 1,025 patient records and 14 clinical attributes. The features include demographics, routine clinical measurements, electrocardiographic results, and exercise-related variables. The target label indicates the presence (1) or absence (0) of heart disease. The class distribution is close to balanced, with 51.32% positive cases. Table I summarizes the features and the selected subset.

B. Data Preprocessing Pipeline

A preprocessing pipeline was used to improve data quality and support stable training. It addresses outliers, missing values, and differences in feature scale.

1) *Outlier Detection and Handling*: Outliers were examined in numerical features, including age, trestbps, chol, thalach, and oldpeak. These variables showed occasional extreme values that could influence model training. The Interquartile Range (IQR) method was used to identify such cases.

The IQR is defined as:

$$\text{IQR} = Q_3 - Q_1 \quad (1)$$

where Q_1 and Q_3 denote the first and third quartiles. Based on this range, lower and upper bounds were computed as:

$$\text{Lower bound} = Q_1 - 1.5 \times \text{IQR} \quad (2)$$

$$\text{Upper bound} = Q_3 + 1.5 \times \text{IQR} \quad (3)$$

Values outside these bounds were clipped rather than removed. This choice preserves all samples while limiting the effect of extreme measurements. The approach reduces the influence of outliers without altering the overall structure of the data.

2) *Feature Engineering*: Feature engineering was employed to create clinically meaningful representations and enhance predictive power.

a) *Categorical Features*: Age was transformed into categorical age groups using binning to capture non-linear relationships with cardiovascular risk:

$$\text{age_group} = \begin{cases} 0, & \text{if age} \leq 40 \\ 1, & \text{if } 40 < \text{age} \leq 50 \\ 2, & \text{if } 50 < \text{age} \leq 60 \\ 3, & \text{if age} > 60 \end{cases} \quad (4)$$

This transformation recognizes that cardiovascular risk increases non-linearly with age, with distinct risk profiles across different life stages.

b) *Interaction Features*: Two interaction features were created to capture synergistic effects between clinical parameters:

$$\text{age_bp} = \text{age} \times \text{trestbps} \quad (5)$$

$$\text{chol_age} = \frac{\text{chol}}{\text{age} + 1} \quad (6)$$

The age_bp feature represents the combined effect of aging and hypertension, while chol_age provides an age-normalized cholesterol measure. These engineered features enhance the model's ability to detect complex relationships between risk factors.

3) *Feature Encoding and Normalization*: Categorical variables were encoded using label encoding, which assigns integer values to categorical levels. All numerical features were standardized using z-score normalization:

$$x_{\text{standardized}} = \frac{x - \mu}{\sigma} \quad (7)$$

where μ represents the mean and σ represents the standard deviation. This standardization ensures all features have zero mean and unit variance, facilitating stable CNN training and convergence.

C. Feature Selection

Feature selection was performed using ANOVA F-values. Features were ranked by their F-statistic, and the top 15 were used for training. The selected set includes the original clinical variables and the engineered features, as shown in Table I. This step reduces dimensionality and helps control training time.

TABLE I: Heart Disease Dataset Features and ANOVA-Selected Features

Feature	Description	Remarks
age	Age in years	Selected
sex	Gender (1 = male; 0 = female)	Selected
cp	Chest pain type (1–4)	Selected
trestbps	Resting blood pressure (mm Hg)	Selected
chol	Serum cholesterol (mg/dl)	Selected
fbs	Fasting blood sugar > 120 mg/dl (1 = true; 0 = false)	Not selected
restecg	Resting electrocardiographic results (0–2)	Selected
thalach	Maximum heart rate achieved	Selected
exang	Exercise induced angina (1 = yes; 0 = no)	Selected
oldpeak	ST depression induced by exercise relative to rest	Selected
slope	Slope of peak exercise ST segment (1–3)	Selected
ca	Number of major vessels colored by fluoroscopy (0–3)	Selected
thal	Thalassemia type (3 = normal; 6 = fixed defect; 7 = reversible defect)	Selected
target	Diagnosis of heart disease (1 = presence; 0 = absence)	Label
age_group	Age category	Engineered
age_bp	Interaction of age and blood pressure	Engineered
chol_age	Cholesterol-to-age ratio	Engineered

D. Model Architecture

1) *1D CNN Design for Tabular Data*: The proposed architecture employs a 1D Convolutional Neural Network specifically designed for tabular data classification. Unlike traditional CNNs for image processing that use 2D convolutions, our approach utilizes 1D convolutions along the feature dimension, treating each patient’s clinical measurements as a 1D signal. The network architecture, summarized in Table II, comprises three convolutional blocks followed by fully connected layers for classification.

TABLE II: CNN Architecture

Layer	Filters/Units	Output Shape	Operations
Input	-	(batch, 1, 15)	-
Conv1D 1	32	(batch, 32, 15)	ReLU, BN, Pool(2)
Conv1D 2	64	(batch, 64, 7)	ReLU, BN, Pool(2)
Conv1D 3	128	(batch, 128, 3)	ReLU, BN, Pool(2)
Flatten	-	(batch, 384)	-
Dense 1	128	(batch, 128)	ReLU, Dropout(0.3)
Dense 2	64	(batch, 64)	ReLU, Dropout(0.3)
Dense 3	2	(batch, 2)	Softmax

The convolutional operation for layer l is defined as:

$$\mathbf{y}_j^l = \sigma \left(\mathbf{b}_j^l + \sum_{i=1}^{N^{l-1}} \mathbf{w}_{ij}^l * \mathbf{x}_i^{l-1} \right) \quad (8)$$

where $\mathbf{y}_j^l$ is the output of the j -th filter in layer l , σ is the ReLU activation function, $\mathbf{b}_j^l$ is the bias term, $\mathbf{w}_{ij}^l$ are the filter weights, $*$ denotes the convolution operation, and $\mathbf{x}_i^{l-1}$ are the inputs from the previous layer.

Batch normalization is applied after each convolutional layer:

$$\hat{x} = \frac{x - \mathbb{E}[x]}{\sqrt{\text{Var}[x] + \epsilon}} \cdot \gamma + \beta \quad (9)$$

where γ and β are learnable parameters, and ϵ is a small constant for numerical stability.

The network employs max pooling with a kernel size of 2 for dimensionality reduction:

$$y_i = \max(x_{2i-1}, x_{2i}) \quad (10)$$

2) *Training Strategy*: The model was trained using the Adam optimizer with an initial learning rate of 0.001. The learning rate was dynamically adjusted using ReduceLROnPlateau scheduler based on validation accuracy with a patience of 5 epochs. Early stopping with a patience of 10 epochs was implemented to prevent overfitting, with a maximum of 100 training epochs.

The loss function employed categorical cross-entropy:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N \sum_{c=1}^C y_{i,c} \log(\hat{y}_{i,c}) \quad (11)$$

where N is the number of samples, C is the number of classes (2 for binary classification), $y_{i,c}$ is the true label, and $\hat{y}_{i,c}$ is the predicted probability. Training utilized mini-batch gradient descent with batch size 32. Dropout regularization of 0.3 was applied to fully connected layers to enhance generalization.

E. Experimental Setup

1) *Data Splitting Strategy*: The dataset contains 1,025 instances. It was split into training and testing sets using a stratified 80-20 split. This resulted in 820 training samples and 205 testing samples. Stratified sampling ensured that both classes were proportionally represented in each set. From the training set, 20% (164 samples) was set aside for validation. This was used for hyperparameter tuning and early stopping. This splitting strategy preserves the original class distribution and allows robust evaluation of model generalization.

2) *Evaluation Metrics*: Comprehensive evaluation metrics were employed to assess model performance from multiple perspectives:

- 1) **Accuracy**: Proportion of correct predictions

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (12)$$

where TP, TN, FP , and FN are true positives, true negatives, false positives, and false negatives, respectively.

- 2) **Precision**: Ability to avoid false positives

$$\text{Precision} = \frac{TP}{TP + FP} \quad (13)$$

- 3) **Recall (Sensitivity)**: Ability to identify true positives

$$\text{Recall} = \frac{TP}{TP + FN} \quad (14)$$

- 4) **F1-Score**: Harmonic mean of precision and recall

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (15)$$

- 5) **AUC-ROC**: Area under the receiver operating characteristic curve, measuring discriminative ability across thresholds

- 6) **Specificity**: Ability to identify true negatives

$$\text{Specificity} = \frac{TN}{TN + FP} \quad (16)$$

- 7) **Confusion Matrix Analysis**: Detailed breakdown of TP, TN, FP , and FN .

All metrics were calculated separately for training and testing sets to assess model generalization and overfitting. The primary evaluation focused on test set performance, with training metrics used to monitor learning progress and convergence.

IV. EXPERIMENTAL RESULTS AND EVALUATION

A. Dataset Statistics

The Heart Disease dataset comprises of 1,025 clinical instances with 14 diagnostic attributes. The dataset exhibits balanced class distribution with 526 positive cases (51.32%) and 499 negative cases (48.68%). Following feature engineering and selection, the dataset was expanded to include 3 engineered features, with 15 optimal features ultimately selected for model training.

B. Model Performance Analysis

1) *Training Phase Results*: On the training set (820 samples), the model achieved 99.76% accuracy with two errors. The confusion matrix contained 420 true positives, 398 true negatives, one false positive, and one false negative. Precision, recall, and F1-score were each 99.76%. The AUC-ROC was 1.000.

2) *Testing Phase Results*: On the test set (205 samples), accuracy was 97.56%. The model produced 103 true positives and 97 true negatives, with three false positives and two false negatives. Precision was 97.17%, recall was 98.10%, and F1-score was 97.63%. The AUC-ROC was 99.80%.

C. Confusion Matrix Analysis

The confusion matrix analysis provides detailed insight into model performance. For the test set:

- 1) **True Positives, TP** : 103 (50.24% of test set) - Correct disease predictions
- 2) **True Negatives, TN** : 97 (47.32% of test set) - Correct healthy predictions
- 3) **False Positives, FP** : 3 (1.46% of test set) - Type I errors (unnecessary interventions)
- 4) **False Negatives, FN** : 2 (0.98% of test set) - Type II errors (missed diagnoses)

The resulting sensitivity (recall) of 98.10% and specificity of 97.00% indicate balanced performance suitable for clinical applications.

D. Comparative Performance Evaluation

1) *Accuracy Metrics Comparison*: Table III presents a comparison of the training and testing phases. The small decrease in performance, with a 2.20% difference in accuracy, indicates that the model generalises well. All measured metrics show consistent performance, with differences between training and testing below 3

TABLE III: Performance Metrics Comparison: Training vs Testing

Metric	Training	Testing	Gap
Accuracy	0.9976	0.9756	0.0220
Precision	0.9976	0.9717	0.0259
Recall	0.9976	0.9810	0.0166
F1-Score	0.9976	0.9763	0.0213
AUC-ROC	1.0000	0.9980	0.0020
Sensitivity	0.9976	0.9810	0.0166
Specificity	0.9975	0.9700	0.0275

2) *Precision-Recall Trade-off*: On the test set, precision was 97.17% and recall was 98.10%. This reflects a low false-positive rate while maintaining a low false-negative rate. The resulting F1-score was 97.63%. Both types of errors are important in the clinical screening for cardiovascular risk, so this balance is useful.

3) *AUC-ROC Analysis*: The model achieved an AUC-ROC of 0.9980 on the test set. This value indicates strong separation between positive and negative cases across a wide range of decision thresholds. In practical settings, the classification threshold may vary depending on whether sensitivity or specificity is prioritized. A high AUC-ROC suggests that the model maintains reliable discrimination even when the operating point is adjusted.

This behavior is relevant to clinical screening tasks, where different risk thresholds may be applied across patient groups or under resource constraints. Rather than reflecting performance at a single cutoff, the AUC-ROC summarizes model behavior across all thresholds. The observed value indicates that the learned feature representations provide a consistent ranking of patients by risk.

E. Overfitting Analysis

The model achieved 99.76% accuracy on the training set and 97.56% accuracy on the test set. The resulting gap of 2.20% suggests that some overfitting is present, but it remains limited. The difference is small enough to indicate that the model does not simply memorize the training data.

Several design choices were made to control overfitting during training. Dropout was applied in the fully connected layers to reduce co-adaptation of neurons. Batch normalization was used after each convolutional layer to stabilize learning dynamics. Early stopping was employed to halt training once validation performance stopped improving. In addition, ANOVA-based feature selection reduced input dimensionality, which further limited model complexity.

Taken together, these measures helped constrain overfitting while preserving high predictive accuracy. The remaining performance gap is expected given the dataset size and model capacity. Overall, the training behavior indicates that the model generalizes reasonably well to unseen data.

F. Feature Importance Analysis

The ANOVA F-value feature selection method identified 15 highly discriminative features. Notably, all three engineered features (*age_group*, *age_bp*, and *chol_age*) were retained. This confirms their clinical relevance and their contribution to heart disease prediction. The selected features span multiple clinical categories. Demographic factors include age, sex, and *age_group*. Clinical measurements include *trestbps*, *chol*, *thalach*, and *oldpeak*. Electrocardiographic indicators consist of *cp*, *restecg*, and *slope*. Exercise-induced responses are represented by *exang*. Advanced diagnostic markers include *ca* and *thal*. Engineered interaction features include *age_bp* and *chol_age*. The inclusion of engineered features alongside traditional clinical variables indicates that the CNN captures both established risk factors and complex feature interactions. This capability improves predictive accuracy. Despite handling diverse feature types, the model remains computationally efficient, with a training time of only 3.54 seconds. This efficiency supports its suitability for clinical environments where rapid decision-making is essential.

V. CONCLUSION AND FUTURE WORK

This paper presented a CNN-based approach for heart disease classification using structured clinical data. On the test set, accuracy reached 97.56% and AUC-ROC reached 99.80%. Sensitivity and specificity were 98.10% and 97.00%, respectively. The training-testing accuracy gap was 2.20%, which indicates limited but non-zero overfitting. Training completed in 3.54 seconds under the reported setup.

The results suggest that combining feature synthesis with a CNN can improve performance on tabular clinical data. The main limitation is the dataset scope. The study uses a single dataset with a fixed feature set and a single split strategy. Future work will evaluate the approach in larger, multicenter cohorts. Additional inputs, including time-series data and imaging, can also be considered if matched datasets are available.

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